

Acoustic Pulses Scattered by a Rigid Sphere Immersed in a Fluid

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The scattered field, produced when a plane-wave pressure pulse is incident upon an ideally rigid sphere, is expressed in the terms of classical scattering theory. Using an incident pulse of known spectral content, the resultant scattered pulse is determined, at a number of field points, with a computer. The time-varying forms of the scattered pulses, thus obtained using the completely rigorous classical theory, display characteristics that may be interpreted in terms of circumferential or creeping waves propagating along the periphery of the sphere. The scattered pulse may be conceived as the superposition of a specularly reflected pulse, with a form similar to the incident pulse, and attenuating pulses that travel along the periphery of the sphere from the boundary of the geometrical shadow to the observation point.

INTRODUCTION

UNDERSTANDING the scattering characteristics of bodies with simple shapes and simple acoustic properties, such as the rigid sphere, is a necessary stepping stone to an adequate understanding of scattering from bodies more complicated in shape, composition, and structure. For the sphere, exact solutions to the scattering problem may be obtained by means of classical analysis in which the scattered field is found by solving the appropriate boundary-value problem, using a Fourier-integral form to synthesize the scattered pulse resulting from a pulse incident upon the body. Aside from their intrinsic interest, the few exact solutions to scattering problems that exist provide useful bench marks in other developments of scattering theory. It is the aim of this paper to demonstrate by numerical computation that the classical theory yields results that may be interpreted in terms of the conceptual model that arises from the creeping-wave formulation of scattering theory.^{1,2} Hickling^{3,4} has shown both the feasibility and the value of applying the classical approach, implemented by use of a computer, to the problem of scattering of acoustic pulses by elastic

spheres and spherical shells. Scattering of a pulse by an ideally rigid sphere is a simpler problem than that of scattering by a sphere whose elastic and structural properties must be taken into account, but this allows investigation of concepts drawn from the creeping-wave formulation of scattering theory^{1,2} unobscured by any effects of the bodies constitution. In a simplified model based on these concepts, the field scattered by a body, which is not excited into either vibration or motion by the incident acoustic field (i.e., the body is fixed and is either acoustically rigid or acoustically soft), is considered to be composed of two types of pulse components. The first type of pulse component is that produced by specular reflection and obeys the laws of geometrical optics. Pulse components of the second type result from creeping waves that are generated at the boundary of the geometrical shadow and travel around the periphery of the sphere, radiating energy as they progress. Since certain surface waves in an elastic body are often also called creeping waves, it is to be emphasized that the creeping waves considered here are waves that travel in the fluid surrounding the rigid scattering sphere. There can be, of course, no elastic surface waves in the case being considered.

I. MATHEMATICAL FORMULATION

In Fig. 1, the center of a rigid sphere, of radius a , is fixed at the origin of a spherical coordinate system. The scattered pressure $p_r(t)$ is to be calculated at the point P in the fluid surrounding the sphere. The scattered pressure at P is the total pressure there, minus the

¹ W. Franz, "Über die Greenschen Funktionen des Zylinders und der Kugel," *Z. Naturforsch.* **9a**, 705-716 (1954).

² H. Überall, R. D. Doolittle, and J. V. McNicholas, "Use of Sound Pulses for a Study of Circumferential Waves," *J. Acoust. Soc. Amer.* **39**, 564-578 (1966).

³ R. Hickling, "Analysis of Echoes from a Solid Elastic Sphere in Water," *J. Acoust. Soc. Amer.* **34**, 1582-1592 (1962).

⁴ R. Hickling, "Analysis of Echoes from a Hollow Metallic Sphere in Water," *J. Acoust. Soc. Amer.* **36**, 1124-1137 (1964).

pressure $p_i(t)$ of the incident wave. Time is denoted by t . The incident wave considered herein is a plane-wave pressure pulse, that is, a plane wave of finite duration, proceeding down the Z axis of a Cartesian coordinate system that also has its origin at the center of the sphere. Consider first, however, not a pulse incident upon the sphere, but an infinitely extended incident plane pressure wave of a single frequency, expressed

$$p_i(t) = p_0 e^{ik(ct-z)}, \quad (1)$$

where p_0 is the pressure amplitude, k is the wave-number (equal to $2\pi/\lambda$, with λ the wavelength), c is the sound speed in the fluid, and z is the Z coordinate. At the point P , this harmonic incident plane wave produces a harmonic scattered wave expressed by the equation

$$p_r(t) = (p_0 a / 2r) e^{ik(ct-r)} f(ka). \quad (2)$$

The expression of Eq. 2 has the form of an outgoing spherical wave modified by $f(ka)$, the reflection form function. The form function depends on the sphere's size parameter ka and also upon the radial and polar spherical coordinates r and θ that locate the point P . (It does not depend upon the coordinate ϕ , due to symmetry of the scattering geometry.) The form function $f(ka)$ thus gives the response of the sphere, as observed at the point P , to a single-frequency incident wave with an arbitrary wavenumber k .

For a transient incident wave, following Kaplan,⁵ the incident pressure pulse can be expressed by a Fourier-integral form

$$p_i(\tau') = \frac{p_0}{2\pi} \operatorname{Re} \int_0^\infty g(ka) e^{ika\tau'} d(ka), \quad (3)$$

where the incident pressure is to be specified by a real function. In Eq. 3, $g(ka)$ is the normalized complex spectrum of the incident plane-wave pulse

$$g(ka) = \frac{2}{p_0} \int_{-\infty}^\infty p_i(\tau') e^{-ika\tau'} d\tau', \quad (4)$$

expressed in terms of the size parameter ka , which is to be considered as a dimensionless frequency. Note that the integral in Eq. 3 is taken only over positive ka values; thus, $g(ka)$ in Eq. 4 is, and need be, defined only for $ka \geq 0$. The quantity τ' is a dimensionless plane-wave propagation parameter, analogous to retarded time, that is defined

$$\tau' = [ct - (z - z_0)]/a, \quad (5)$$

where z_0 is that Z coordinate at which the phase of the incident plane wave is zero at $t=0$. Henceforth, $z_0=0$, so the phase of the incident wave will be zero at the

⁵ W. Kaplan, *Operational Methods for Linear Systems* (Addison-Wesley Publ. Co., Reading, Mass., 1962), pp. 235-238.

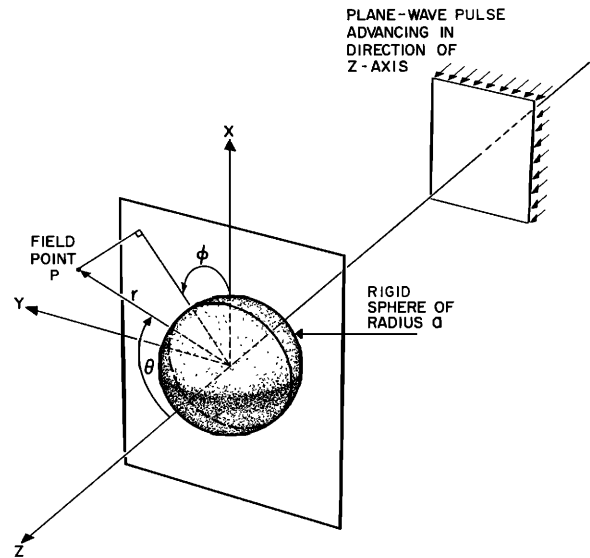


FIG. 1. Geometry in the classical formulation of the scattering problem.

center of the sphere. Physically, Eq. 3 expresses the incident plane-wave pulse $p_i(\tau')$ as the superposition of differential plane harmonic waves, each with a frequency ka and a relative amplitude and phase determined by the pertinent value of the pulse spectrum $g(ka)$. Just as the harmonic plane wave of Eq. 1 produced the harmonic spherical wave of Eq. 2, each such differential plane harmonic wave produces, on encountering the sphere, a differential harmonic spherical wave. The scattered spherical-wave pulse $p_r(\tau)$, observed at P , is the superposition of all such differential scattered waves and is given by the Fourier-integral form

$$p_r(\tau) = \left(\frac{p_0 a}{2r} \right) \left[\frac{1}{2\pi} \operatorname{Re} \int_0^\infty g(ka) f(ka) e^{ika\tau} d(ka) \right], \quad (6)$$

where

$$\tau = (ct - r)/a \quad (7)$$

is a propagation parameter appropriate to an outgoing spherical wave. If, in Eq. 7, one writes

$$\Psi(\tau) = \frac{1}{2\pi} \operatorname{Re} \int_0^\infty g(ka) f(ka) e^{ika\tau} d(ka), \quad (8)$$

then by computing $\Psi(\tau)$ as a function of τ , the time-varying form of the scattered pulse observed at P will be obtained. That is, an observer at P would see, as physical time t increased, a pulseform develop like that computed according to Eq. 8 for increasing values of the parameter τ . However, expressions for the reflection form function $f(ka)$ and the incident-pulse spectrum $g(ka)$ must be obtained before Eq. 8 can be evaluated.

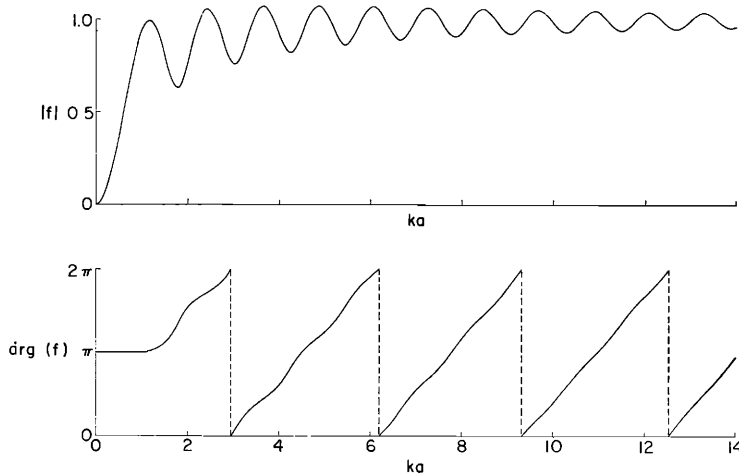


FIG. 2. Reflection form function $f(ka)$ versus sphere size parameter ka for monostatic scattering observed in the farfield.

The expression for the form function is obtained⁶ by solving the appropriate boundary-value problem, using the method of separation of variables and imposing Neumann boundary conditions at the sphere's surface. The result is that

$$f(ka) = -2 \left(\frac{kr}{ka} \right) e^{ikr} \sum_{n=0}^{\infty} (-i)^{n+1} h_n^{(2)}(kr) \times P_n(\cos\theta) (2n+1) \sin\eta_n e^{i\eta_n}. \quad (9)$$

In this expression, $h_n^{(2)}(kr)$ is the n th order spherical Bessel function of the third kind, defined

$$h_n^{(2)}(x) = j_n(x) - iy_n(x),$$

with $j_n(x)$ and $y_n(x)$, respectively, the n th-order spherical Bessel functions of the first and second kind. In Eq. 9, $P_n(\cos\theta)$ is the n th-order Legendre polynomial, with the cosine of the polar angle θ as argument, and η_n is obtained using the expression

$$\tan\eta_n = - \frac{[(n+1)j_n(ka) - (ka)j_{n-1}(ka)]}{[(n+1)y_n(ka) - (ka)y_{n-1}(ka)]}. \quad (10)$$

Note by the presence of r and θ in Eq. 9 that $f(ka)$ depends on the location of P . Thus, for each particular point P at which a scattered pulse form is to be computed, a distinct evaluation of Eq. 9 must be inserted in Eq. 8. For P in the farfield, an asymptotic expression for $h_n^{(2)}(kr)$ is used, eliminating the dependence of $f(ka)$ on the coordinate r . In the farfield then,

$$f(ka) = - \frac{2}{ka} \sum_{n=0}^{\infty} P_n(\cos\theta) (2n+1) \sin\eta_n e^{i\eta_n}. \quad (11)$$

Figure 2 shows a plot of the complex form function over

a small range of ka , for monostatic backscattering with P in the farfield. Note in Fig. 2 the undulations that occur in $|f(ka)|$ as ka is varied. One can find a plot of this type in Stenzel.⁷

If the form of the incident pressure pulse $p_i(\tau')$ is specified, the pulse spectrum $g(ka)$ is found using Eq. 4. In this work, all incident pulseforms are truncated sinusoids beginning at $\tau' = 0$. That is,

$$p_i(\tau') = p_0 \Pi[(\tau'/m) - \frac{1}{2}] \sin(k_0 a \tau'), \quad (12)$$

where

$$\Pi(x) = \begin{cases} 1, & |x| \leq \frac{1}{2} \\ 0, & |x| > \frac{1}{2} \end{cases}$$

and m is the length of the pulse expressed nondimensionally. The quantity $k_0 a$ is the size-parameter value that relates the frequency of the incident truncated sinusoid to the radius of the sphere. Note that if the reflecting sphere were not present, the leading edge of the incident pulse would arrive at the origin at $\tau' = 0$ and that the leading edge of the pulse encounters the sphere (at the point $z = a$, $\theta = \pi$) when $\tau' = -1$. The length of an incident pulse having b sinusoidal cycles is fixed by the relation

$$m = 2b\pi(k_0 a)^{-1}. \quad (13)$$

For incident pulses that are truncated sinusoids, $g(ka)$ is expressed

$$g(ka) = \alpha^{-1}(1 - e^{-i\alpha}) - \beta^{-1}(1 - e^{-i\beta}), \quad (14)$$

where

$$\alpha = (ka + k_0 a),$$

and

$$\beta = (ka - k_0 a).$$

⁶ S. N. Rachevkin, *A Course of Lectures on the Theory of Sound* (The Macmillan Co., New York, 1963), pp. 350-353.

⁷ H. Stenzel, *Guide for the Calculation of Sound Processes*, English transl. (Verlag Julius Springer, Berlin, 1939).

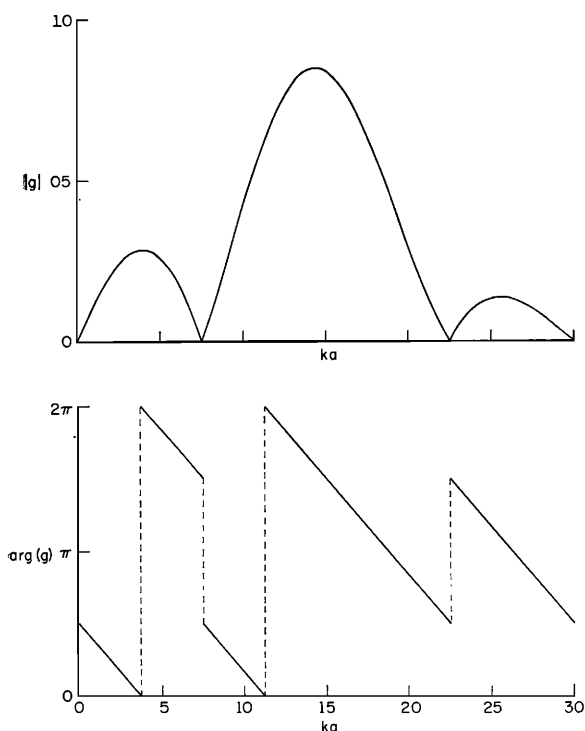


FIG. 3. Spectrum $g(ka)$ of a 2-cycle pulse, with $k_0a=15.0$, plotted versus sphere size parameter ka .

The limiting form of Eq. 14 is used when $ka=k_0a$. In Fig. 3, the incident-pulse spectrum is plotted over a small range of ka , for a $b=2$ cycle pulse with $k_0a=15.0$.

II. NUMERICAL CONSIDERATIONS

The integration over ka in Eq. 8 is accomplished using the trapezoidal rule, inserting into the integrand each of a number of discrete values of τ , with the interval between successive τ values chosen small enough to allow the variation of $\Psi(\tau)$ to be adequately displayed. The size of the increments taken in ka , when computing $\Psi(\tau)$ for a given value of τ , was 0.001π . Values of $f(ka)^8$ and $g(ka)$ were initially computed in such size-parameter increments, and the results stored on magnetic tape; these tapes were subsequently used as inputs to the integration program. Thus, an integration, yielding $\Psi(\tau)$ for a particular value of τ , involved 32 000 complex values of the integrand.⁹ Trigonometric-function values, needed in evaluating the exponential term in the integrand, were obtained by cyclically selecting entries from a precomputed, 5000-entry, table of the sine, and

⁸ A. J. Rudgers, "Techniques for Numerically Evaluating the Formulas Describing Monostatic Reflection of Acoustic Waves by Elastic Spheres," NRL Rep. 6551 (1967).

⁹ For the rigid sphere problem, a somewhat coarser increment in ka would have reduced cumulative computation error. However, this enhanced error has no significant effect upon the results presented and in view of other uses of the particular computer programs involved, it was of little value in this to employ a coarser incrementation interval.

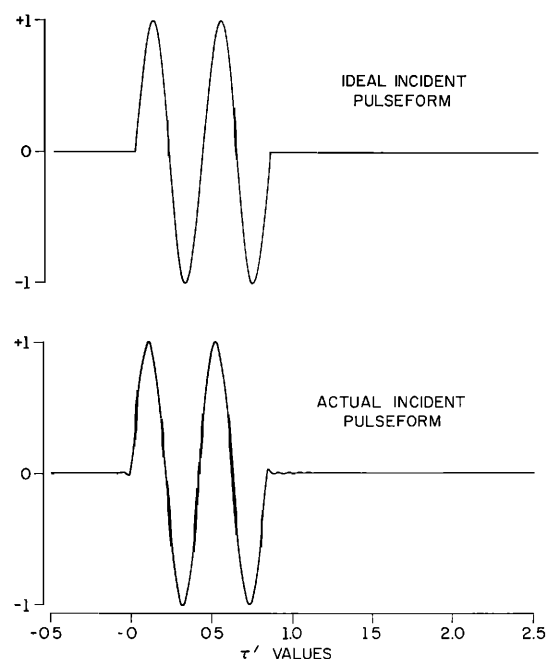


FIG. 4. Comparison of ideal and actual incident pulseforms for a two-cycle pulse with $k_0a=15.0$.

interpolating linearly between the selected entries. The linear interpolation was also carried out by cyclically selecting tabular entries. Both cyclical selection operations, used in order to avoid the prohibitively time-consuming computation of large numbers of trigonometric-function values, were synchronized to the incrementation of ka in the trapezoidal-rule evaluation of the integral of Eq. 8.

Replacing the integral's infinite upper limit in Eq. 8 with a finite value has the effect of passing the incident pulse, in this case a truncated sinusoid, through an ideal low-pass filter with a cutoff at that ka value chosen as the upper limit of the integral. Therefore, the actual incident pulseform is a filtered replica of the ideal pulseform given by Eq. 12. That is, limiting the range of integration in Eq. 8 at some maximum ka value is equivalent to choosing an incident pulseform that has no frequency components above this maximum ka value. In all cases of computed pulseforms, 32π was the upper limit taken for the integral in Eq. 8. Figure 4 presents a comparison of an ideal two-cycle-long truncated-sinusoid pulse, having $k_0a=15.0$, with the actual incident pulse, when the limit of the integral in Eq. 8 (and, therefore, the cutoff frequency of the ideal low-pass filter) is taken at $ka=32\pi$.

The numerical method used to compute reflected pulseforms was checked by determining the pulseform scattered by a hypothetical body for which the form function $f(ka)\equiv 1$ for all values of the size parameter. This effects an assembly of the incident pulseform by means of the numerical technique. However, with $f(ka)\equiv 1$, the integral of Eq. 8, with a finite upper

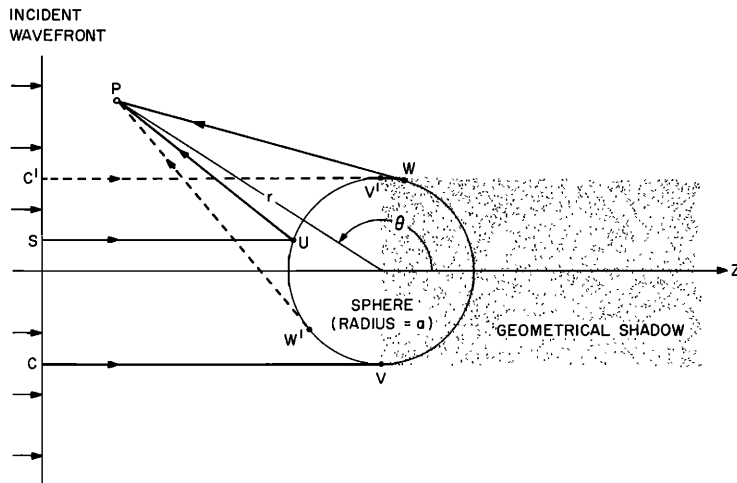


FIG. 5. Geometry of the model used for interpreting scattered pulseforms.

limit, can also be expressed in terms of sine-integral and cosine-integral functions. A pulseform expressed in this way was evaluated and then compared with the pulseform resulting from the numerical assembly procedure. This comparison showed that the numerical method yielded correctly computed values of $\Psi(\tau)$, with an error small enough to not significantly affect the validity of the results presented, even when $\Psi(\tau)$ assumes small values.

III. INTERPRETING SCATTERED PULSEFORMS

Before showing examples of scattered pulseforms computed according to Eq. 8, a simple conceptual model of scattering by a rigid sphere is described, which incorporates the most important predictions of the creeping-wave formulation of scattering theory.^{1,2} This model is only an approximate picture of the scattering phenomena, but one that will be useful later when discussing the computed pulseforms that are the exact representation of the scattered field. The model is as follows. Consider that the scattered pulse observed at P results from the superposition of a number of what will be called "pulse components." Except that it can be of a different amplitude, each pulse component is to be considered a replica of the incident pulse. Various pulse components arrive at the observation point P along geometrical ray paths as depicted in Fig. 5, where, since the scattered field is symmetrical with respect to the coordinate ϕ of Fig. 1, P can be taken in the plane of the sphere's section defined by the XZ plane. There are two cases to be considered, depending on whether P is located within or outside the sphere's geometrical shadow—this shadow region being the semi-infinite cylinder, of radius a , having its base in the XY plane (Fig. 5).

When P is outside the shadow, there are two types of pulse components whose superposition results in the scattered pulse observed at P : (1) a pulse component of the specularly reflected type and (2) a number of

pulse components of the creeping-wave type. Two sequences of pulse components of the creeping-wave type arrive at P , in each sequence, there being a number of pulse components that arrive at P at periodic intervals. The first creeping-wave-type pulse-component arrival at P is due to a wave that travels with speed c straight along the ray CV in Fig. 5 to the edge of the geometrical shadow, proceeds with speed c^* by a ray path VW that lies along the sphere's periphery, and finally travels to P with speed c along the path WP that is tangent to the sphere's surface. The second creeping-wave-type pulse component arrives at a later time, after traveling with speed c along path $C'V'$, thence, from point V' , around the sphere with speed c^* to point W' , from which it proceeds to P , along the tangent $W'P$, with speed c . These two pulse components are, respectively, the first member in each of the two sequences of creeping-wave-type pulse components previously mentioned. The other creeping-wave components in these sequences will arrive at P at later times. Such later arrivals are the creeping-wave-type pulse components that travel completely around the sphere one or more times before arriving at P . That is, the first creeping-wave-type pulse component previously mentioned arrives at P from W , but not all the peripherally traveling energy arriving at W is radiated to P . A portion of it continues on around the sphere's surface by a path through successive points V' , U , W' , and V , eventually arriving at W again. Here, a portion is radiated to P and a portion continues on, making successive appearances at W , at which times energy is radiated to P , giving rise to a sequence of creeping-wave-type pulse-component arrivals there. A similar sequence of pulse-component arrivals begins with the other creeping-wave-type pulse component, which started at C' . Suppose that a particular feature of the incident pulseform (e.g., the second positive peak) would arrive at the coordinate origin at time t_0 if the scattering sphere were absent. Then the corresponding

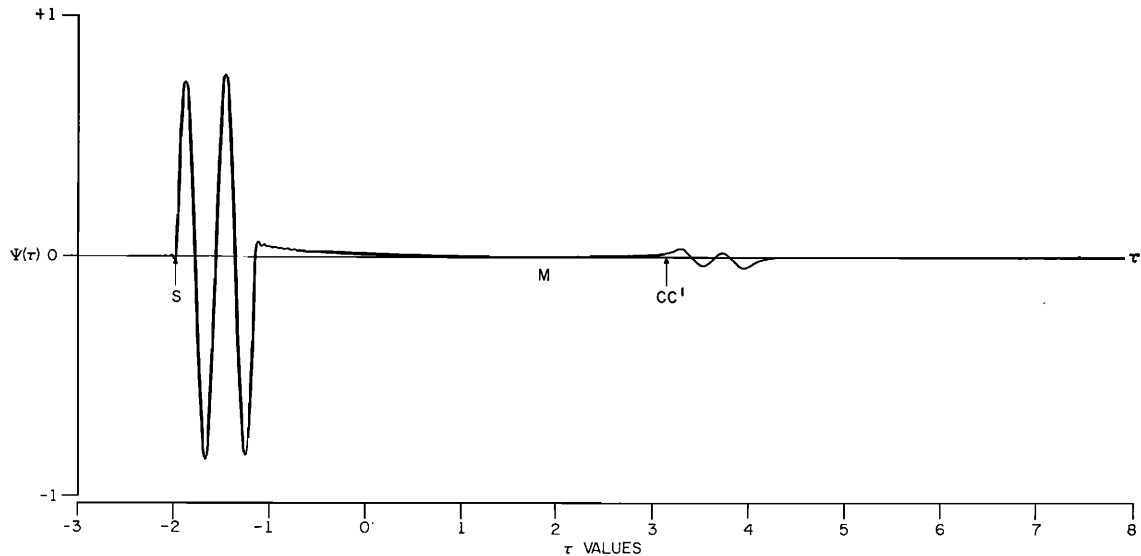


FIG. 6. Scattered pulseform computed for monostatic backscattering into the farfield (see Fig. 7), with a two-cycle-long incident pulse having $k_0 a = 15.0$.

feature in the creeping-wave-type pulse components will arrive at point P at values of the propagation parameter τ given by

$$\tau_{1,2} = (ct_0/a) + (c/c^*)[\gamma + (2N+1)\pi \mp (\pi - \theta)] + \Lambda, \quad (15)$$

where $\gamma = \arcsin(a/r)$ and

$$\Lambda = [(r/a)^2 - 1]^{1/2} - (r/a).$$

In Eq. 15, and in Eq. 16 to follow, the subscripts 1 and 2 of τ pertain respectively to each of the two creeping-wave-type pulse-component sequences; the $-$ of the double sign goes with the subscript 1 and the $+$ with the subscript 2. The creeping-wave-type pulse component makes N complete circuits of the sphere's periphery before it arrives at P . For the first pair of arrivals, one from each sequence, $N=0$; for the second $N=1$, etc. As mentioned, c^* is the speed of creeping-wave-type pulse components along the sphere's periphery, and Eq. 15 applies when the observation point P is located outside the geometrical shadow of the sphere. For the other possible case, with P in the shadow, the formula

$$\tau_{1,2} = (ct_0/a) + (c/c^*)[\gamma + 2N\pi \mp \theta] + \Lambda \quad (16)$$

gives the values of τ at which a feature of a creeping-wave-type pulse component, corresponding to some feature of the incident pulse, is observed to arrive at P when it travels by creeping-wave ray paths.

In addition to pulse components of the creeping-wave type, there is a specularly-reflected type pulse component in the scattered field, which arrives at P by path SUP in Fig. 5 according to the laws of geometrical optics, if P is not located in the shadow. The specularly reflected-type pulse component arrives before any creeping-wave-type pulse component. It is difficult to

find a simple expression that gives the value of τ for which a particular pulse feature arrives at a general field point P via the path of specular reflection. For P located far from the sphere, however, (strictly, $r \rightarrow \infty$) a feature of the pulse component of the specularly reflected type arrives at P , by the ray path of specular reflection, at a value of τ given by

$$\tau_s = (ct_0/a) - 2 \sin(\frac{1}{2}\theta) \quad (17)$$

while, if P lies on the surface of the sphere, one has

$$\tau_s = (ct_0/a) + (\cos\theta - 1). \quad (18)$$

Additional remarks about the picture of scattering just described are necessary. First, as mentioned before, this picture is not a complete description of the predictions of the creeping-wave formulation of scattering theory. From the creeping-wave formulation, however, comes the notion of distinct specularly reflected and peripherally traveling waves, which can be considered to propagate along certain geometrical ray paths.

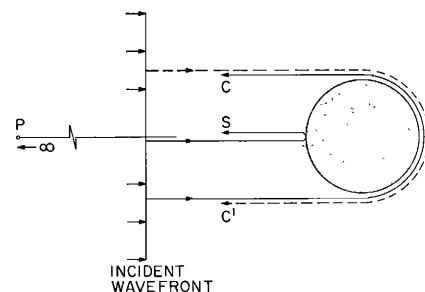


FIG. 7. Geometry for which pulse of Fig. 6 was computed ($r \rightarrow \infty$, $\theta = \pi$), showing ray paths of the scattered-pulse components.

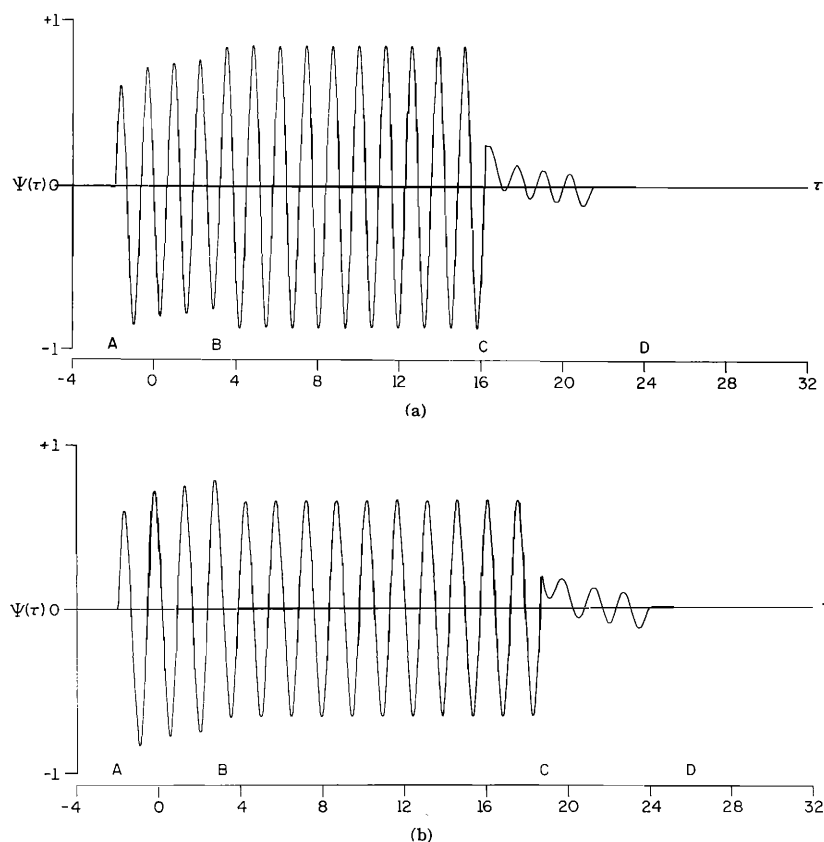


FIG. 8. Interference of specularly reflected-type and creeping-wave-type pulse components in scattered pulses. Incident pulses are 14 cycles long in each case. (a) Scattered pulseform for constructive interference, $k_0a = 4.851$. (b) Scattered pulseform for destructive interference, $k_0a = 4.238$.

Franz¹ showed that the ray paths that are important for the scattering by a sphere are quite similar to those that are important in scattering by a cylinder. For a sphere, to a first approximation, as for a cylinder, only waves that arrive at the observation point P by paths following a geodesic curve on the surface make a significant contribution to the creeping-wave-type pulse component there, so that the significant ray paths shown in Fig. 5 look essentially like those that would be found if the scattering body were a cylinder rather than a sphere. Thus, there is a difference between cases where P lies on the Z axis and cases where it does not; in the former situation, an ensemble of rays, one originating at each point on the sphere's great-circle shadow-boundary, arrive at P , while in the latter situation only two rays are important. To answer the question, however, "How does the ensemble of rays get sorted into just two rays as we move P off axis?" requires further investigation.

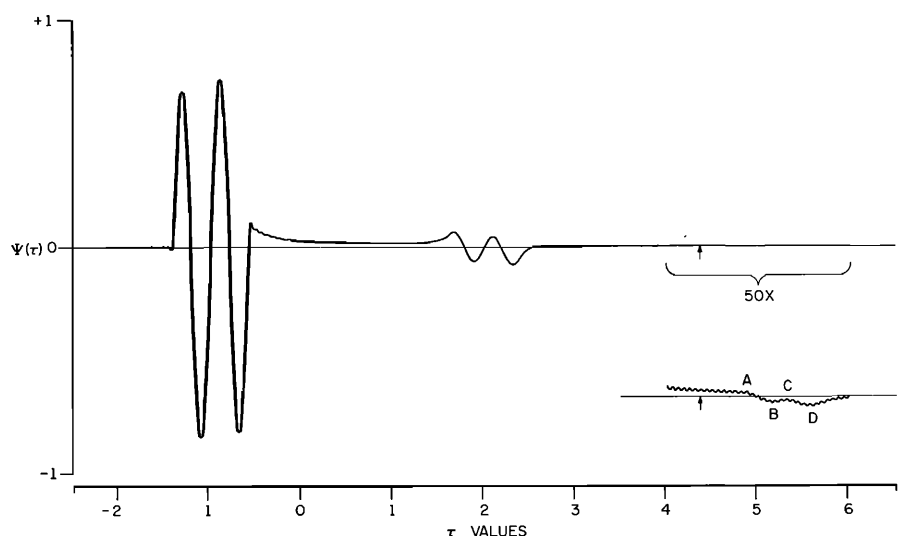
Second, propagation effects, which make pulse components inexact replicas of the incident pulse are not accounted for in the model described, nor does it predict the speed and attenuation of pulse components as they propagate along the various ray paths. Nevertheless, in the precisely computed scattered pulseforms to be shown, pulse components will be found that correspond to pulse components of the types described, and

these will resemble the incident pulse. Thus, while the model predicts pulse components of the types that are found to make up a scattered pulse, it is too simplified to fully account for all details observed in the exact scattered pulseforms obtained from evaluation of Eq. 8.

IV. SCATTERED PULSES

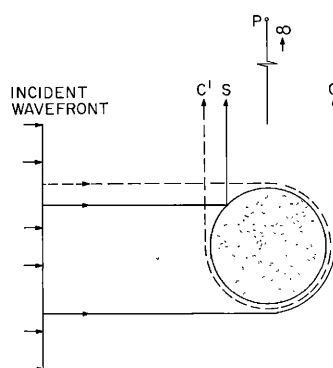
Figure 6 displays a scattered pulse that is computed according to Eq. 8 for monostatic backscattering in the farfield ($\theta = \pi$, $r \rightarrow \infty$, see Fig. 7). The incident pulse is a two-cycle truncated sinusoid with $k_0a = 15.0$ and its pulselength is $m = 4\pi/15$, or approximately 0.838 times the sphere's radius. The two pulse components obvious in the scattered pulse are explained by the previously described model. The first component of the scattered pulse in Fig. 6, denoted by S , is a pulse component of the specularly-reflected type and, according to Eq. 17, arrives at P at $\tau = -2$, traveling by ray path S as depicted in Fig. 7. The smaller pulse component in Fig. 6, denoted by CC' , is due to the ensemble of creeping waves that simultaneously arrive at P . These waves start from all points on the great-circle boundary of the geometrical shadow and then proceed half-way around the sphere's periphery by paths such as C and C' shown in Fig. 7. For the pulse component generated by these waves, $N=0$ and $\tau_1 = \tau_2$ in Eq. 15.

Fig. 9. Scattered pulseform computed for scattering into the farfield at $\theta = \pi/2$ (see Fig. 10), with a two-cycle-long incident pulse having $k_0 a = 15.0$.



The creeping-wave-type pulse component in question arrives at a value of $\tau \approx \pi$. Although distortion of the creeping-wave-type pulse component makes the determination somewhat imprecise, the ratio of propagation speeds c^*/c can be found by noting the values of τ at which particular features of the creeping-wave-type pulse component are observed. Such a determination gives $c^*/c = 0.97$ at $k_0 a = 15.0$. That is, in this case, peripherally traveling waves traverse the sphere's surface at a (group) speed approximately equal to, but slightly less than, the free-field sound speed in the fluid. At $k_0 a = 15.0$, the creeping-wave-type pulse component of the monostatically scattered pulse, observed in the farfield, has an average amplitude that is 4.3% of the amplitude of the specularly reflected-type pulse component. The value of the attenuation, calculated for the creeping waves, is 1.27 Np/rad. The creeping-wave-type pulse component observed is that for which $N = 0$ in Eq. 15; the creeping-wave-type pulse components having $N > 0$ are so low in amplitude and so highly attenuated that they cannot be detected in the presence of Gibbs-phenomenonlike oscillations about the τ axis and noise-like error introduced by the computing process.

Fig. 10. Geometry for which pulse of Fig. 9 was computed ($r \rightarrow \infty$, $\theta = \pi/2$), showing ray paths of scattered-pulse components.



In this case, and in subsequent cases considered, the exact scattered pulseforms have more complicated characteristics than those that are expected from the previously described model. First, as will be noted in Fig. 6, the specularly reflected-type pulse component is not a precise replica of the incident pulseform. Not only are the pulse peaks of slightly different amplitudes, but the specularly reflected-type pulse component possesses a long tail. In Fig. 6, this tail decays from its peak above the τ axis at $\tau = -1.14$ to a minimum at $\tau = M$. The entire tail is above the τ axis and, from the minimum at M , there is a steady buildup in the amplitude of the scattered pulse, culminating in the creeping-wave-type pulse component at $\tau = CC'$. Thus, between the occurrences of the specularly reflected-type and the creeping-wave-type pulse components, there exists a small positive contribution to the scattered field. Creeping-wave-type pulse components, although having pulseforms clearly similar to those of the incident pulses, are also found in all cases to be distorted—lacking sharp onsets and terminations, and having positive and negative peaks of unequal amplitudes that are not symmetrically disposed about the τ axis.

Computations by Rheinstein,¹⁰ based on classical theory, have previously demonstrated the occurrence of separated specularly-reflected-type and creeping-wave-type pulse components in electromagnetic scattering. Earlier than Rheinstein, Hickling³ declared the initial portion of the pulse scattered by an elastic sphere to be due to specular reflection—this portion often appearing in his computed pulseforms as a replica of the incident pulse, separate from the remainder of the scattered pulse. However, this specular contribution to the scattered return by an elastic sphere, was taken by Hickling to be the entire response of a rigid spherical

¹⁰ J. Rheinstein, "Backscatter from Spheres: A Short Pulse View," IEEE Trans. Antennas Propagation AP-16, 89-97 (1968).

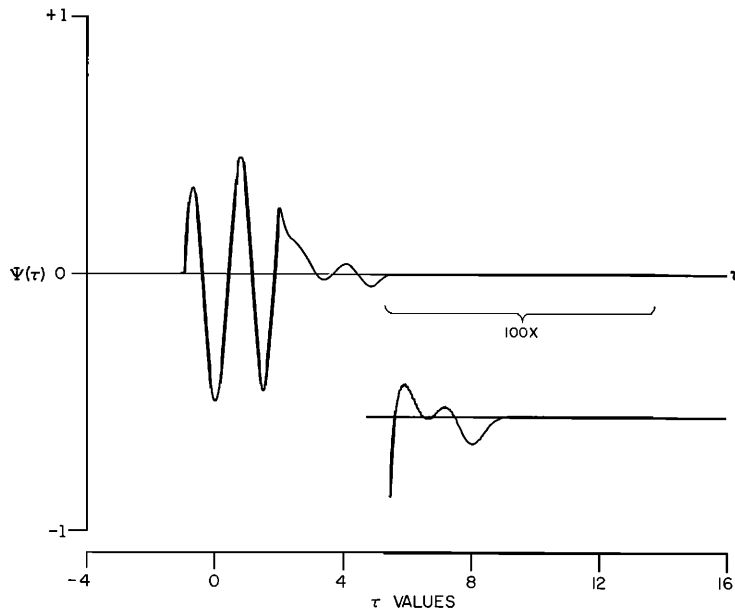


FIG. 11. Scattered pulseform computed for the observation point P on the sphere's surface at the shadow boundary (see Fig. 12), with a two-cycle-long incident pulse having $k_0a = 4.238$.

scatterer. The portion of the scattered pulse not associated with the specular arrival he attributed to elastic behavior of the scatterer,¹¹ and thus his interpretation overlooked the contributions of creeping-wave-type pulse components (of the rigid-body or Franz type) to the scattered field of a rigid sphere. The contributions of the creeping waves were, of course, present in Hickling's exactly computed results, but were effectively masked by the presence of the more pronounced elastic effects.

That a pulse component is observed near $\tau = \pi$ in Fig. 6, determines that the component arose due to waves originating at the shadow boundary and traveling around behind the sphere. The "image-pulse" theory of Freedman^{12,13} also predicts the observation of separate pulse components, but provides an inadequate model for describing the pulse components shown in Fig. 6. According to the image-pulse model, not only should the pulse component beginning at $\tau = -2$ be observed, but a second pulse should arrive at $\tau = 0$, having originated by backscattering from the shadow boundary. The absence of such an image-pulse and the presence in Fig. 6 of a pulse component at $\tau \approx \pi$, for which there is no provision in the image-pulse model, show that the image-pulse method does not predict the occurrence of the components that arise out of the exact theory of scattering by a rigid sphere.

Interference will occur between the specularly reflected-type and creeping-wave-type pulse components

¹¹ K. J. Diercks and R. Hickling, "Echoes from Hollow Aluminum Spheres in Water," *J. Acoust. Soc. Amer.* **41**, 380-393 (1967).

¹² A. Freedman, "A Mechanism of Acoustic Echo Formation," *Acustica* **12**, 10-21 (1962).

¹³ A. Freedman, "The High Frequency Echo Structure of Some Simple Body Shapes," *Acustica* **12**, 61-70 (1962).

if the incident pulselength is sufficiently great. Figures 8(a) and 8(b) show scattered pulses, computed for the monostatic farfield geometry of Fig. 7, with 14-cycle-long incident truncated-sinusoid pulses, having $k_0a = 4.851$ and $k_0a = 4.238$, respectively. In each scattered pulse of Fig. 8, for values of τ between A and B , the specularly reflected-type pulse component alone arrives at the observation point, while for τ values between C and D , the scattered pulse is due primarily to the arrival of the creeping-wave-type pulse component. Between B and C , however, pulse components overlap and interference occurs between the end of the specularly reflected-type pulse component and the beginning of the creeping-wave-type pulse component, giving rise to regions of steady sinusoidal oscillations. The amplitude of the oscillations in such regions corresponds closely to the ordinate of the $|f(ka)|$ -vs- ka curve of Fig. 2 at the value of ka equal to the value of k_0a that pertains to the incident pulse. That is, the amplitude of the steady oscillation that occurs in the region

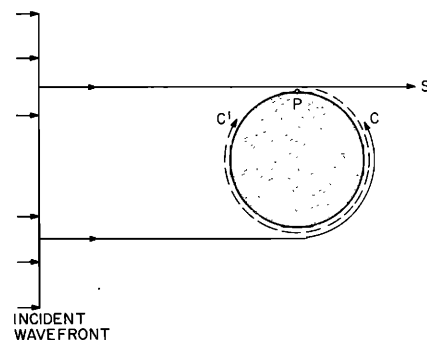
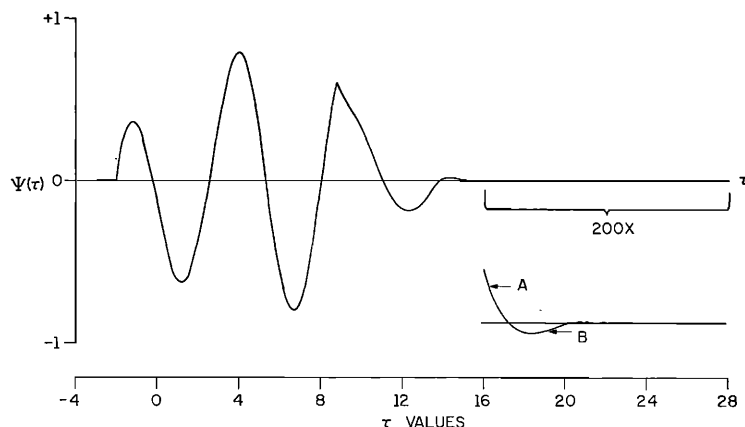


FIG. 12. Geometry for which the pulse of Fig. 11 was computed ($r = a$, $\theta = \pi/2$), showing ray paths of scattered-pulse components.

FIG. 13. Scattered pulseform showing the $N=1$ creeping-wave-type pulse component. Monostatic backscattering into the farfield (see Fig. 7), with a two-cycle-long incident pulse having $k_0a=1.166$. In the insert, the magnified tail of the $N=0$ creeping-wave-type pulse component is labeled *A* and the last half-cycle of the $N=1$ creeping-wave-type pulse component is labeled *B*.



$B < \tau < C$ of a scattered pulse is almost exactly the same as the amplitude of the scattered harmonic wave that is produced when a single-frequency infinitely extended plane wave is incident on the sphere, if ka for the incident harmonic wave is the same as k_0a for the incident pulse. Thus, the steady-state scattering characteristics of the sphere, as described by the undulating form-function-magnitude curve of Fig. 2, are produced primarily by interference of the specularly-reflected-type pulse component and the initial creeping-wave-type pulse components that are present in the scattered pulse. Interference is constructive when the specularly-reflected- and creeping-wave-type pulse component add in-phase and destructive when they add out-of-phase. Figure 8(a) shows constructive interference occurring when k_0a for the incident pulse is taken at an abscissa of the curve of Fig. 2 for which a maximum exists in the modulus of the form function ($k_0a=4.851$). Destructive interference, shown in Fig. 8(b), occurs when k_0a corresponds to the abscissa of a minimum in this $|f(ka)|$ -vs- ka curve ($k_0a=4.238$). Note also from Fig. 2 that the effect of the creeping-wave-type component must diminish as ka increases, since the variation between maxima and minima of $|f(ka)|$ becomes less and less.

Figure 9 displays the scattered pulse computed for P located in the farfield at an angle $\theta=\pi/2$. The incident pulse is again a 2-cycle truncated sinusoid with $k_0a=15.0$. Two prominent pulse components are again present in the scattered pulse, which are to be expected according to the previously discussed model. The first, arriving at $\tau \approx -\sqrt{2}$, is the specularly reflected-type pulse component that traveled by path S as depicted in Fig. 10. The second pulse component, arriving at $\tau \approx \pi/2$, traveled around one-quarter of the sphere's circumference by path C . According to the model, there is also a third scattered-pulse component in this asymmetrical scattering geometry, which travels by the path C' of Fig. 10 around three-quarters of the sphere's circumference and arrives at P at $\tau \approx 3\pi/2$. This component is not prominent in Fig. 9 due to the rapid attenuation of the peripherally traveling pulses. How-

ever, plotting $\Psi(\tau)$, in the τ interval bracketed in Fig. 9, with its ordinate scale magnified 50 times, renders this creeping-wave-type pulse component visible. The component is seen, in the insert of Fig. 9, as a much-distorted 2-cycle pulse on which a higher-frequency undulation¹⁴ rides. Its maxima and minima at *A*, *B*, *C*, and *D* are apparent.

Figure 11 shows a scattered pulse computed for P located on the sphere's surface at the boundary of the geometrical shadow ($r=a$, $\theta=\pi/2$). A 2-cycle pulse with $k_0a=4.238$ is incident. The incident pulse length at this frequency is 2.965 times the sphere's radius and there is incomplete separation of the scattered pulse components. Again there is the specularly reflected-type pulse component, beginning at $\tau \approx -1$ in Fig. 11, which is the disturbance created as the incident pulse sweeps by the point P along path S , as shown in Fig. 12. On the tail of the specularly reflected-type pulse component, the creeping-wave-type pulse component arrives at $\tau \approx \pi$. This component traveled half-way around the sphere by path C of Fig. 12. Plotting the bracketed region of the τ axis, with the ordinate magnified 100 times (insert of Fig. 11), makes visible the creeping-wave-type pulse component that traveled by path C' of Fig. 12 completely around the sphere's circumference, arriving at P at $\tau \approx 2\pi$. From the τ interval between the arrivals of the second positive peak of the specularly reflected-type and of the creeping-wave-type pulse component, a ratio of speeds $c^*/c=0.96$ is found. This value is derived by applying either Eq. 16 or Eqs. 15 and 18, since in this case the creeping-wave-type pulse component, launched at the shadow boundary, appears first as the specularly reflected-type pulse-component disturbance created at the shadow boundary on the opposite side of the sphere. Other ways of reckoning

¹⁴ High-frequency low-amplitude undulation in the scattered pulseforms is due to the presence of undulation at the head and tail of the incident pulseform (Fig. 4). Undulation in the incident pulseform is a manifestation of the Gibbs phenomenon. Always present, it is least prominent when the spectrum of the incident pulseform has low amplitude at the higher ka values for the range of integration in Eq. 8.

the ratio c^*/c yield slightly different values. Creeping-wave-type pulse components that travel more than once around the sphere (i.e., those for which $N > 0$) are again not visible due to attenuation. It is seen, however, that creeping-wave-type pulse components truly manifest themselves at the surface of the scattering sphere and the same types of pulse components occur at the surface that were previously found in the farfield.

At small values of k_0a , the creeping-wave-type pulse component that traveled $1\frac{1}{2}$ times around the sphere may be discerned (for this, $N=1$ in Eq. 15). Figure 13 shows the scattered pulse computed, with the monostatic farfield geometry of Fig. 7, for an incident 2-cycle pulse having $k_0a=1.166$ (this value corresponding to the abscissa of the first maximum value of $|f(ka)|$ in Fig. 2). By Eq. 13, the incident pulselength is fixed and the various pulse components overlap for this value of k_0a . At the beginning of the scattered pulse, prior to $\tau \approx \pi$, the specularly reflected-type pulse component alone arrives at P . After this, the creeping-wave-type pulse component, which has traveled half-way around the sphere, begins to arrive, interfering with the specularly reflected-type pulse component. Although the tail of the specularly reflected-type pulse component obscures the shape of the creeping-wave-type pulse component, the last half-cycle and tail of this $N=0$ creeping-wave-type pulse component can be seen $11.1 < \tau < 16.0$. However, if the bracketed portion of the τ axis, containing part of the tail of the $N=0$ creeping-wave-type pulse component, is plotted with the ordinate scale magnified 200 times (insert of Fig. 13) then the last half-cycle of the $N=1$ creeping-wave-type pulse component (labeled B) is seen in the region $17.2 < \tau < 20.3$.

Thus, it is possible in this case to detect the presence of the second creeping-wave-type pulse component, the one traveling $1\frac{1}{2}$ times around the sphere, which is predicted by creeping-wave theory. From the computed results at $k_0a=1.166$, the attenuation of creeping waves is found to be 1.01 Np/rad. Calculation in this case, and in the case of the $k_0a=1.166$ sphere with P located on its surface at an angle $\theta=\pi/2$, indicate the group speed of the creeping-wave-type pulse component to be 4% to 6% greater than that of a free wave in the fluid, but pulseform distortion and the absence of pulse separation make this speed determination imprecise.

V. CONCLUSION

The pulse scattered by a rigid sphere may be precisely computed at any desired field point according to the rigorous classical theory. A simplified model that draws on the ideas of the creeping-wave formulation of scattering theory predicts the observed occurrence of a number of components in such scattered pulses. The exact computations verify the picture that results from the creeping-wave formulation of scattering theory, as indeed they must, since the physical content in the creeping-wave and the classical formulations of scattering theory are identical.

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